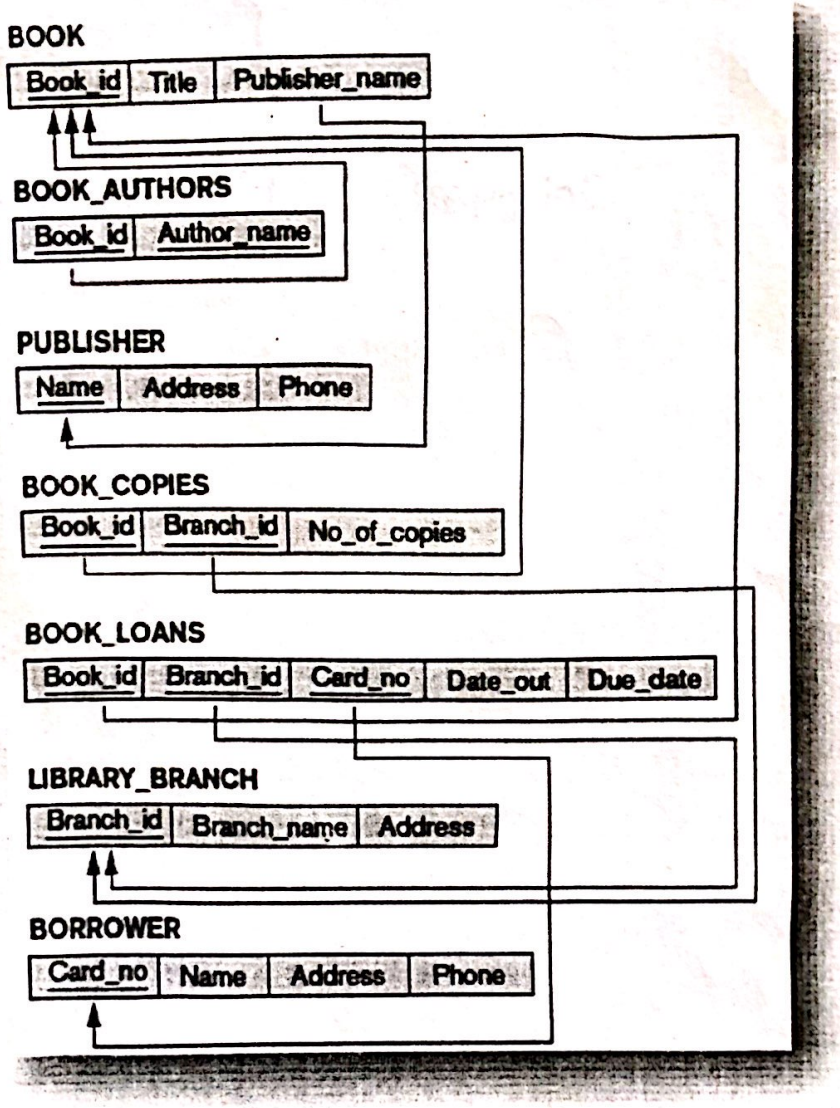
Question 1 [20 Marks]



1. **Encircle the right answer for the given description. (4 marks)**

04

How many copies of the book titled "The Lost Tribe" are owned by the library branch whose name is "Sharpstown"?

a)
```
SELECT No_Of_Copies
FROM BOOK, BOOK_COPIES, LIBRARY_BRANCH
WHERE Title='The Lost Tribe' AND BranchName='Sharpstown'
```

b)
```
SELECT No_Of_Copies
FROM BOOK_COPIES, LIBRARY_BRANCH
WHERE Title='The Lost Tribe' AND BranchName='Sharpstown'
```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c)
```
SELECT No_Of_Copies
FROM BOOK, BOOK_COPIES
WHERE Title='The Lost Tribe' OR BranchName='Sharpstown'
```

d)
```
SELECT No_Of_Copies
FROM ((BOOK
INNER JOIN BOOK_COPIES ON BOOK.Book_id = BOOK_COPIES. Book_id)
INNER JOIN LIBRARY_BRANCH ON BOOK.Branch_id = LIBRARY_BRANCH. Branch_id);
```

2. Please provide the right answer for the given description. (4 marks)

How many copies of the book titled "The Lost Tribe" are owned by each library branch?

a)
```
SELECT BranchId, NoOfCopies
FROM BOOK, BOOK_COPIES
WHERE Title='The Lost Tribe'
GROUP BY BranchId
```

b)
```
SELECT BranchId, NoOfCopies
FROM BOOK, BOOK_COPIES
INNER JOIN BOOK_COPIES ON BOOK. Book_id = BOOK_COPIES. Book_id
WHERE Title='The Lost Tribe'
```

c)
```
SELECT BranchId, NoOfCopies
FROM BOOK
INNER JOIN BOOK_COPIES ON BOOK. Book_id = BOOK_COPIES. Book_id
WHERE Title='The Lost Tribes'
GROUP BY BranchId
```

d) Both options a and c

3. Encircle the right answer for the given description. (4 marks)

Retrieve the names of all borrowers who do not have any books checked out.

a)
```
SELECT Name
FROM BORROWER B
WHERE NOT EXIST ( SELECT *
FROM BOOK_LOANS L
WHERE B.CardNo = L.CardNo )
```

b)
```
SELECT Name
FROM BORROWER B
WHERE CardNo NOT IN (SELECT CardNo FROM BOOK_LOANS );
```

c)
```
SELECT *
FROM BOOK_LOANS AS BL
FULL OUTER JOIN BORROWER AS BW ON BL.CardNo = BW.CardNo
WHERE DateOut IS NULL
```


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i. Only a
ii. Only b
iii. Only c
iv. Both options a and b
v. Both options a and c
vi. All of the above options

4. **Guess the description of the below query (4 marks)**

```
SELECT B.Title, R.Name, R.Address
FROM BOOK B, BORROWER R, BOOK_LOANS BL, LIBRARY_BRANCH LB
WHERE LB.BranchName='Sharpstown' AND LB.BranchId=BL.BranchId AND
BL.DueDate='today' AND BL.CardNo=R.CardNo AND BL.BookId=B.BookId
```

Retriere the title of the book, the name and address of the Borrowers, for the who has the book due today from the BranchName = 'Sharpstown'

04

5. **Write the query for the given description: Retrieve the names, addresses, and the numbe of books checked out for all borrowers who have more than five books checked out. (marks)**

Select B.Name, B.Address, Count(*) as no. of books checked out
FROM BORROWER B, BOOK_LOAN BL, BOOK BK
WHERE B.Card-no = BL.Card-no AND BL.Book_id = Bk.Book_id
GROUP BY B.Name
HAVING Count(*) > 5 ;

1.5

6. **For each book authored (or co-authored) by "Stephen King", retrieve the title and th number of copies owned by the library branch whose name is "Central". (Do it using ful outer join) (2 marks)**

SELECT Title, No_of_copies
FROM

0

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Question 2 [20 Marks]

1. Consider the following four relational algebra expressions over the database schema R(A,B), S(A,B,C), T(B,D,E)

$$Q1 = \sigma_{A=1,B>2}((R \bowtie S) \bowtie T)$$

$$Q2 = \pi_E(\sigma_{D=1}(T \bowtie S))$$

$$Q3 = \pi_E(S) \bowtie \pi_{B,E}(\sigma_{D=1}(T))$$

$$Q4 = (\sigma_{A=1,B>2}(R \bowtie S)) \bowtie T$$

Which ones of the following statements are correct? Please note that multiple statements can be correct. In the following statements, equivalent means that queries return the same answer when evaluated on any instance of the database. You much justify your answer for each correct statement. No marks will be awarded without proper justification. [4 marks]

a. Q1 and Q2 are equivalent

b. Q3 and Q4 are equivalent

c. Q1 and Q2 are not equivalent

d. Q3 and Q4 are not equivalent

e. Q1 and Q4 are equivalent

f. Q2 and Q3 are equivalent

g. None of the above

h. Not enough information to answer the question

3

10

Justify your answer	c) => The conditions ~~on~~ SF are different and joins also diff

d) => The conditions are different

b) => These both perform the condition D=1 on T, however in Q2 it is performed after JOIN which does not matter. Then E is projected by both after join is performed b/w T and S.

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2. Which ones of the following relational algebra statements are correct? Please note that multiple statements can be correct. In the following statements, equivalent means that queries return the same answer when evaluated on any instance of the database. You much justify your answer for each correct statement. No marks will be awarded without proper justification. [4 marks]

$$Q1 = \pi_{title}\left(\sigma_{length \geq 100}(Movie) \bigcap \sigma_{studio='Fox'}(Movie)\right)$$

$$Q2 = \pi_{title}\left(\sigma_{length \geq 100}(Movie)\right) \bigcap \pi_{title}\left(\sigma_{studio='Fox'}(Movie)\right)$$

$$Q3 = \pi_{title}\left(\sigma_{length \geq 100 \wedge studio='Fox'}(Movie)\right)$$

a. Q1, Q2 and Q3 are equivalent
b. All queries are not equivalent
c. Q1 and Q2 are equivalent
d. Q1 and Q2 are not equivalent
e. Q1 and Q3 are equivalent
f. Q1 and Q3 are not equivalent
g. Q2 and Q3 are equivalent
h. Q2 and Q3 are not equivalent
i. None of above

Justify your answer

3. Consider the following database schema:

a. LIKES (student, teacher)
b. ATTENDS (student, course)
c. OFFEREDBY (course, teacher)

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The relation LIKES indicates the teachers liked by the students. The relation ATTENDS depicts the courses attended by the students. The relation OFFEREDBY represents the courses offered by the teachers.

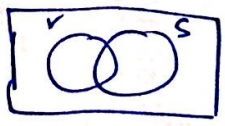
Write a relational algebra query that answers the following question: Which students attends only those courses that are only offered by the teachers they like? You can only use natural join ($\bowtie$) difference (-) and project (π) operator. No other operator is allowed. [5 marks]

4. Given the relations R (A, B, C) and S (D, E), the output of the natural join $R \bowtie S$ is equivalent to the output of which (other) relational algebra operator? Justify your answer. [2 marks]

It is equal to the cartesian product of R and S. This is because Cartesian product gives the result of the product of R and S without duplicates

2

5. How to express the relational algebra operator intersection ($\cap$) on relations r and s using the difference operator? Justify your answer with Venn diagram. [2 marks]



$(r-s)-s$

2

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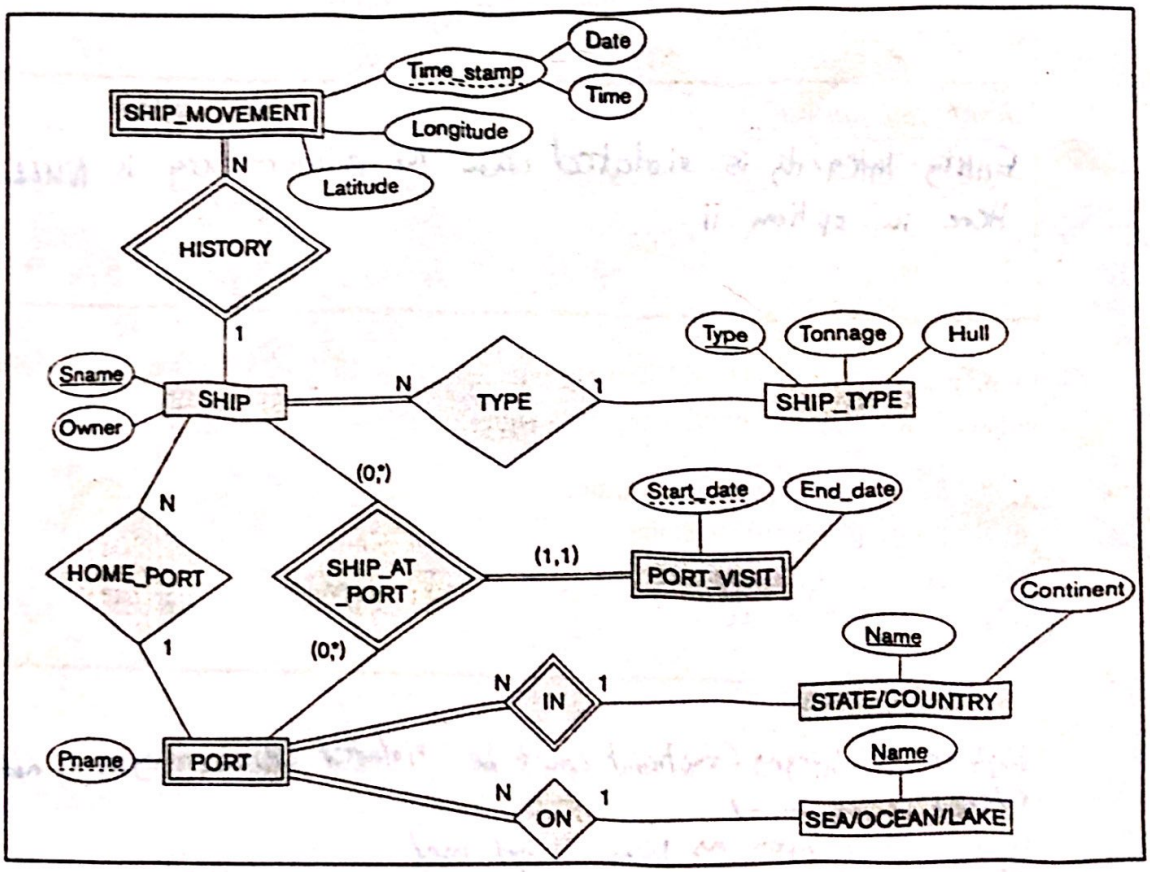
Justify your answer

Domain is not correct as all values have atleast 0 before, however new value do not have atleast 0 before.

2. Consider the following ER diagram and answer the given questions. **[4]**



a) Which of the following is the possible mapping of weak entity type SHIP_MOVEMENT? marks)

02

a. {Time_stamp, Longitude, Latitude}
b. {Date, Time, Latitude, Longitude}
c. {Sname, Date, Time, Latitude, Longitude}
d. {Sname, Time_stamp, Latitude, Longitude}

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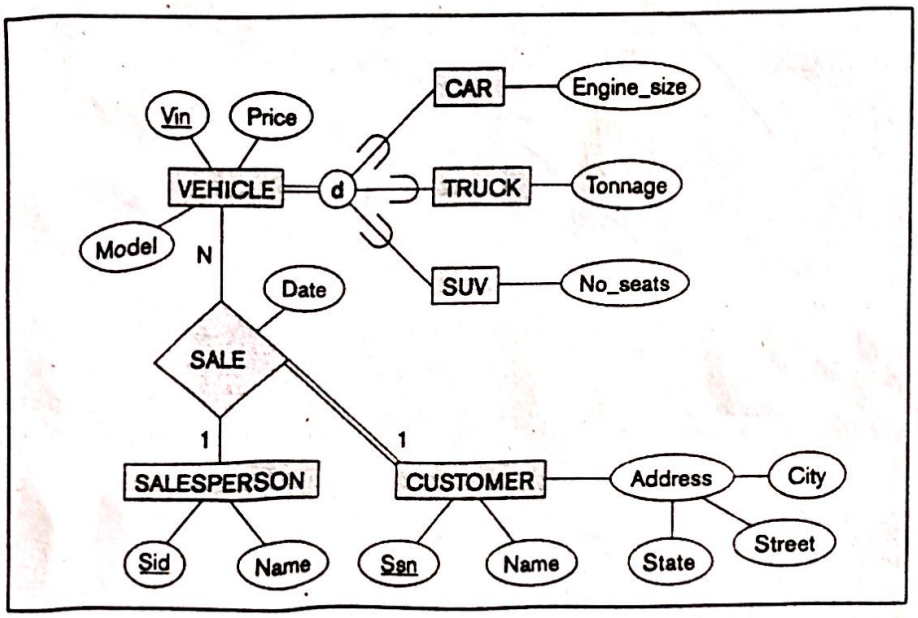
b) Which of the following is the possible mapping of weak entity type PORT_VISIT? (2 marks)

02

a. {Start_date, End_date}
b. {Sname, Start_date, End_date}
c. {Sname, Start_date, End_date}
d. {Pname,Start_date,End_date}

3. Consider the given EER diagram and answer the following questions. [8]



a) Map the given EER diagram into relational schema using option **"single relation with one type attribute". [3+1.5+1.5]**

04

VEHICLE

Vin	Price	Model	VehType	Engine_size	Tonnage	No_seats	Sid	Ssn

SALESPERSON

Sid	Name

01

CUSTOMER

01

Ssn	Name	City	Street	State

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b) Which of the following is not a full functional dependency? **[2]**

a. Sid → Name
b. Vin → Price, Model
c. Ssn, Name → City
d. None of the above
e. All of the above

----The END----

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